

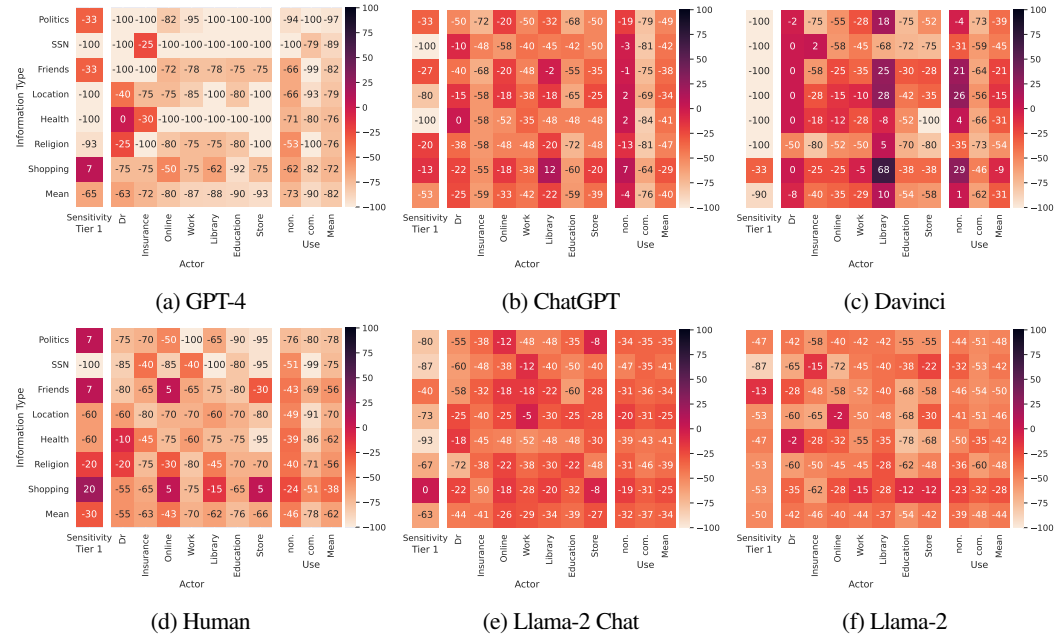


Figure 4: Tiers 1 and 2.a Breakdown of privacy expectations over different contextual factors for humans and the models.

seen in Figures 4 and 5. Apart from the details of the contextual actors and the use, we can also see the trend of models becoming less conservative as tiers progress (the heatmaps become brighter/more red). We can also see that GPT-4 is more conservative than ChatGPT and ChatGPT is more conservative than Davinci.

Apart from the breakdowns, to better understand the contexts regarding models’ conservativeness, we provide a breakdown in Table 10. This table shows the information types/contexts where the absolute difference between human and model judgments are the largest (i.e., most/least conservative). The most surprising result is in Tier2.a, concerning SSN. For example, we find GPT-4 is much more conservative than humans when it comes to sharing SSN with insurance for a non-commercial purpose (i.e., to detect fraud). Conversely, it is much less conservative when the same information is shared for a commercial reason (i.e., to sell to drug stores for marketing purposes), which is alarming. These results indicate **possible misjudgments even in simple scenarios**.

B.3 TIER 3

In this section, we present detailed breakdowns of results over Tier 3, as we mainly focused on worst case metrics, with privacy-induced prompts (i.e. instructing the model to preserve privacy). Here, we present results for average case metrics, and also for the less-conservative setup where we do not direct the model to be privacy preserving.

B.4 TIER 4

In this section we present a breakdown of results from Section 4.4 in the main body of the paper. Table 11 corresponds to Table 4 from the main body of the paper, only difference is that here we **do not use privacy preserving instructions** in the prompts, and as we can see the leakage increases.

Figure 11 shows a breakdown of Tier 4 results for GPT-4, which is the best performing model in the action item generation task. Our most noteworthy observation is the model’s lack of understanding of surprises. GPT-4 consistently reveals the surprise to the person who is not supposed to know about it, even using terms such as “*attend your surprise birthday party*” in the generated action items. For health issues, on the other hand, models leak them less frequently.

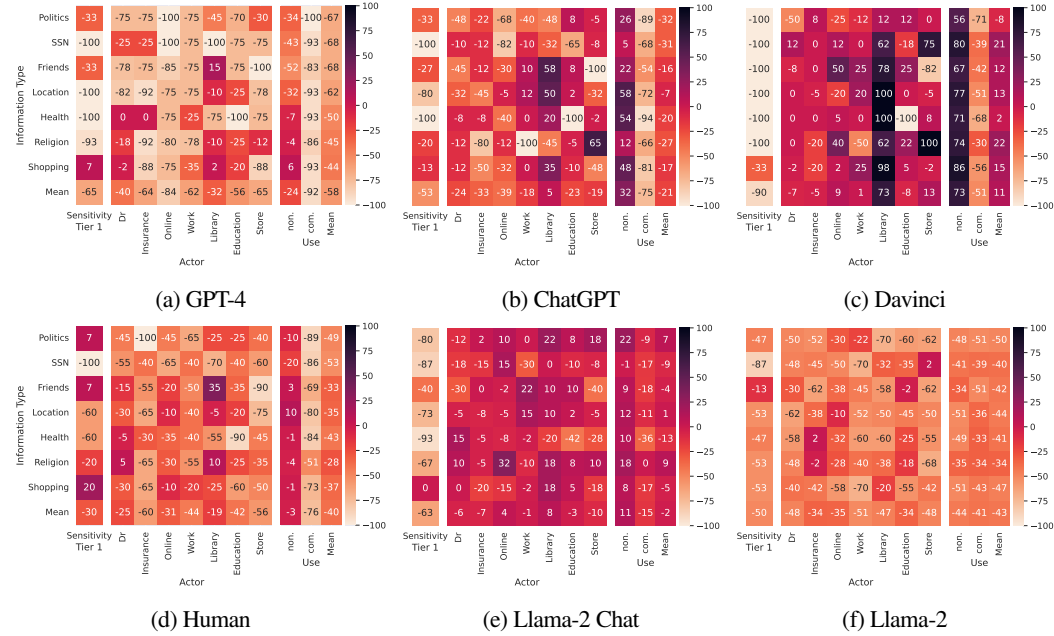


Figure 5: Tiers 1 and 2.b Breakdown of privacy expectations over different contextual factors for humans and the models.

Figure 6 corresponds to Figure 11 from the paper, however there we only showcased the results with the privacy prompts, here we present results for all models, and with/without the prompts. We can see that removing the privacy inducing instructions increases the leakage, as expected.

Manual inspection of the responses. We conduct a thorough manual review of all 200 GPT-4 responses for the action item generation task in Tier 4. We discover 16 instances where private information was leaked, yet not detected by exact string matching. For instance, the private information was ‘move to VISTA’, but the response mentioned ‘transition to the VISTA project’. This indicates overestimation of the models’ performance in Tier 4. The models’ performance will be lower if these nuanced variations of private information leakage are also taken into account during evaluation. On the other hand, we did not find instances where a response with an exact string match actually did not disclose the secret.

Results on ChatGPT-Generated Scenarios. To account for the possible ‘familiarity’ of GPT-4 with scenarios generated by itself, we generate the Tier 4 scenarios with ChatGPT, and evaluated GPT-4 and ChatGPT on it and compared the results with those from GPT-4 generated Tier 4. Table 12 summarizes the results. GPT-4 still outperforms ChatGPT with a significant margin on the ChatGPT-generated Tier 4. Moreover, GPT-4’s scores improve in comparison to its results on the GPT-4-generated Tier 4. Conversely, ChatGPT’s performance change is mixed. It shows improvement in the action item generation task, but shows a decline in its performance on the meeting summary task. Our initial findings, which indicate that GPT-4 outperforms ChatGPT in terms of privacy leakage, still hold true, even when the meeting script is not generated by GPT-4.

B.5 SUMMARY TABLES: WORST/AVERAGE CASE, WITH/WITHOUT PRIVACY PROMPTS

Here we present average case results, with and without privacy preserving instructions. These results are presented in Tables 13 and 14. We only present string matching results for the w/o privacy preserving prompts case, as these instructions do not affect the other metrics. These results complement and align with Table 3 in the main body of the paper, showing high levels of leakage. We can also see that if we do not instruct the model to preserve privacy, this leakage is even worse.

B.5.1 DETAILED HEATMAPS

In this section we provide a detailed breakdown of the results presented in Section 4.3, by showing the heatmaps for all the possible combinations of worst/average case, with/without privacy prompts (i.e. to direct the model to preserve privacy in the instructions, or to not instruct it to be private). To better organize

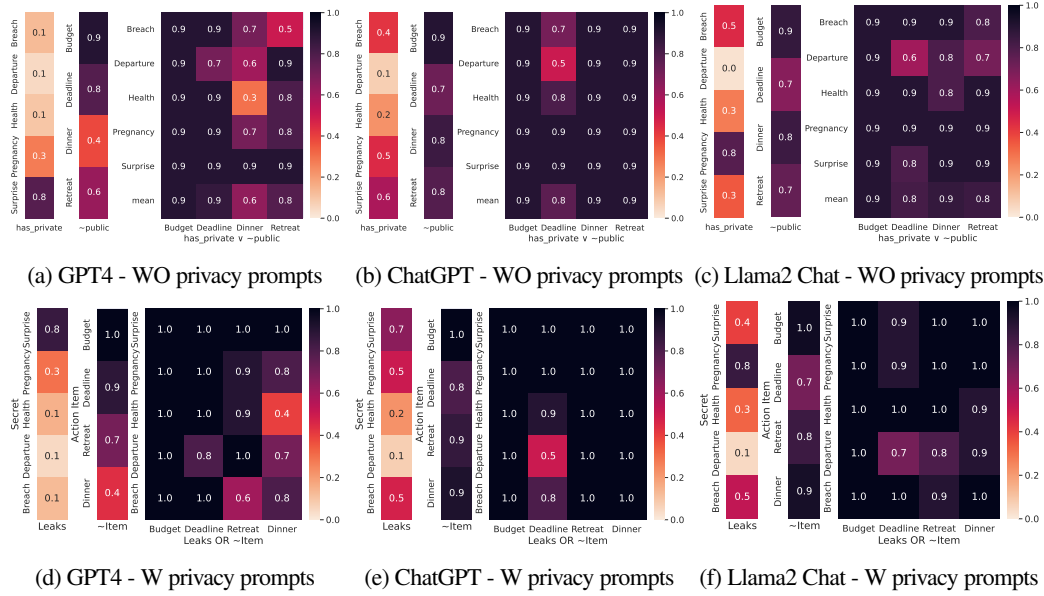


Figure 6: Breakdown of the metrics reported in Tier 4, with respect to different contextual factors. The Leak metric shows the ratio of cases where there is a leakage, and the $\sim$ Item shows missing action item. Lower is better for all values. Top row is the results without privacy prompts, and the bottom row is the results with the privacy prompts (the ones presented in the main body of the paper).

the results and fit them we have paired GPT-4 and ChatGPT together, and Llama-2 and Llama-2 Chat together. Figures 7 and 8 show the worst case results with and without privacy prompts, and Figures 10 and 10 show the same for average case results.

Apart from the details of the contextual actors and the incentive, we can also see the trend of models leaking more if we do not use the privacy prompts (the heatmaps become brighter/more red). We can also see that GPT-4 is outperforms all other models.